Angles of elevation and depression

ANSWER KEY

Core Concepts: Angles of elevation and depression

1 In $\triangle ABC$ studying angles of elevation and depression, $m\angle A = 52^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 83° ✓

b. 93°

c. 75°

d. 128°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (52^\circ + 45^\circ) = 83^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 15

c. 13 ✓

d. 18

Substituting $x = 1$ into $y = 5x + 8$ gives $y = 5(1) + 8 = 13$.

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying angles of elevation and depression, $m\angle A = 52^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 76° ✓

b. 86°

c. 68°

d. 128°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (52^\circ + 52^\circ) = 76^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 13

b. 20

c. 18 ✓

d. 23

Substituting $x = 2$ into $y = 5x + 8$ gives $y = 5(2) + 8 = 18$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying angles of elevation and depression, $m\angle A = 52^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 69° ✓

b. 79°

c. 61°

d. 128°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (52^\circ + 59^\circ) = 69^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm ✓

c. 30 cm

d. 96 cm

Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?

a. 18

b. 25

c. 23 ✓

d. 28

Substituting $x = 3$ into $y = 5x + 8$ gives $y = 5(3) + 8 = 23$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying angles of elevation and depression, $m\angle A = 52^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 62° ✓

b. 72°

c. 54°

d. 128°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (52^\circ + 66^\circ) = 62^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 23

b. 30

c. 28 ✓

d. 33

Substituting $x = 4$ into $y = 5x + 8$ gives $y = 5(4) + 8 = 28$.

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